

SHETH LUJ COLLEGE

AI PRACITCAL - 2

Naive Bayes' Classifier

- Implement the Naive Bayes' algorithm for classification.
- Train a Naive Bayes' model using a given dataset and calculate class probabilities.
- Evaluate the accuracy of the model on test data and analyze the results.

Output



```
Python 3.13.7 (tags/v3.13.7:bcee1c3, Aug 14 2025, 14:15:11) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Daniyal khan/AI_PRACTICAL/naive1.py =====
Predicted Test Results : ['Iris-virginica' 'Iris-versicolor' 'Iris-setosa' 'Iris-virginica'
'Iris-setosa' 'Iris-virginica' 'Iris-setosa' 'Iris-versicolor'
'Iris-versicolor' 'Iris-versicolor' 'Iris-virginica' 'Iris-versicolor'
'Iris-versicolor' 'Iris-versicolor' 'Iris-versicolor' 'Iris-setosa'
'Iris-versicolor' 'Iris-versicolor' 'Iris-setosa' 'Iris-setosa'
'Iris-virginica' 'Iris-versicolor' 'Iris-setosa' 'Iris-setosa'
'Iris-virginica' 'Iris-setosa' 'Iris-setosa' 'Iris-versicolor'
'Iris-versicolor' 'Iris-setosa']

Model Accuracy : 100.0 %

Model Confusion Matrix :
[[11 0 0]
 [ 0 13 0]
 [ 0 0 6]]

>>>
```

Diabeteese

Output

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Enter "help" below or click "Help" above for more information.

>>> ===== RESTART: C:/Daniyal khan/AI_PRACTICAL/naive2.py =====
Libraries imported
Dataset Loaded
Pregnancies Glucose BloodPressure ... DiabetesPedigreeFunction Age Outcome
0 6 148 72 ... 0.627 50 1
1 1 85 66 ... 0.351 31 0
2 8 183 64 ... 0.672 32 1
3 1 89 66 ... 0.167 21 0
4 0 137 40 ... 2.288 33 1

[5 rows x 9 columns]
Transformed Data
[[ 0.63994726  0.84832379  0.14964075 ... 0.20401277  0.46849198
  1.4259954 ]
 [-0.84488505 -1.12339636 -0.16054575 ... -0.68442195 -0.36506078
 -0.19067191]
 [ 1.23388019  1.94372388 -0.26394125 ... -1.10325546  0.60439732
 -0.10558415]
 ...
 [ 0.3429808  0.00330087  0.14964075 ... -0.73518964 -0.68519336
 -0.27575966]
 [-0.84488505  0.1597866 -0.47073225 ... -0.24020459 -0.37110101
  1.17073215]
 [-0.84488505 -0.8730192  0.04624525 ... -0.20212881 -0.47378505
 -0.87137393]]

Model:
```

```
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-0.27575966]
[-0.84488505  0.1597866 -0.47073225 ... -0.24020459 -0.37110101
 1.17073215]
[-0.84488505 -0.8730192  0.04624525 ... -0.20212881 -0.47378505
 -0.87137393]]

Model:
GaussianNB()

Predicted Values:
[0 1 1 0 0 0 0 1 0 1 0 1 1 1 0 0 0 0 1 0 1 1 0 0 0 1 1 0 0 0 0 1 0 0 0
 0 1 1 1 1 0 0 0 0 1 1 0 1 0 0 0 0 1 1 0 0 0 0 0 1 0 1 1 0 0 1 1 0 1 1 0 0
 1 0 0 0 0 0 0 1 0 0 0 1 0 1 0 1 1 1 1 0 0 1 0 0 1 0 0 1 0 0 1 0 0 1 1 0
 1 0 0 1 0 0 0 1 1 0 0 0 0 0 1 0 1 0 0 0 1 0 0 0 0 0 0 0 1 0 0 1 1 0 1 0 0
 1 1 0 1 0 0 0 0 1 0 0 1 1 0 0 0 1 0 0 0 0 1 0 1 1 0 0 0 1 1 1 0 0 0 0 0 0
 0 0 0 1 1 1 1 1 0 0 1 0 0 0 1 0 1 1 1 0 0 0 1 0 1 1 0 0 0 0 1 1 0 0 0 1
 1 0 0 1 0 0 1 0 0]

Confusion Matrix:
[[116  31]
 [ 29  55]]

Accuracy: 0.7402597402597403
Model Score: 0.7402597402597403

Prediction Probabilities:
Squeezed text (231 lines).

ROC AUC: 0.7973760932944606
```

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Accuracy: 0.7402597402597403
Model Score: 0.7402597402597403

Prediction Probabilities:
[[9.86158573e-01 1.38414272e-02]
 [3.12380377e-02 9.68761962e-01]
 [5.40635227e-02 9.45936477e-01]
 [9.65065125e-01 3.49348746e-02]
 [7.50736253e-01 2.49263747e-01]
 [6.17355267e-01 3.82644733e-01]
 [9.73541945e-01 2.64580550e-02]
 [9.49535030e-01 5.04649697e-02]
 [1.97506776e-03 9.98024932e-01]
 [9.33568191e-01 6.64318089e-02]
 [3.96675687e-02 9.60332431e-01]
 [9.25723305e-01 7.42766952e-02]
 [4.43087076e-02 9.55691292e-01]
 [3.24714802e-01 6.75285198e-01]
 [4.19899828e-01 5.80100172e-01]
 [7.58424433e-01 2.41575567e-01]
 [8.25176629e-01 1.74823371e-01]
 [9.43192727e-01 5.68072733e-02]
 [8.52330066e-01 1.47669934e-01]
 [9.41638552e-01 5.83614478e-02]
 [2.14629528e-01 7.85370472e-01]
 [8.07450257e-01 1.92549743e-01]
 [7.64915233e-03 9.92350848e-01]
 [6.77931736e-09 9.99999993e-01]
 [9.65206667e-01 3.47933331e-02]
 [9.62938073e-01 3.70619271e-02]
```

```
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[[9.17095570e-01 8.29044299e-02]
 [4.65788173e-01 5.34211827e-01]
 [1.34472113e-02 9.86552789e-01]
 [7.57267043e-01 2.42732957e-01]
 [2.89299832e-03 9.97107002e-01]
 [3.55108946e-02 9.64489105e-01]
 [8.91247444e-01 1.08752556e-01]
 [8.13243646e-01 1.86756354e-01]
 [4.33692506e-01 5.66307494e-01]
 [8.11073619e-01 1.88926381e-01]
 [8.21611917e-01 1.78388083e-01]
 [9.55690889e-01 4.43091110e-02]
 [9.95634774e-01 4.36522619e-03]
 [8.65709747e-01 1.34290253e-01]
 [9.50399535e-01 4.96004645e-02]
 [8.80658610e-01 1.19341390e-01]
 [2.95626173e-02 9.70437383e-01]
 [9.65190563e-01 3.48094365e-02]
 [9.32730774e-01 6.72692264e-02]
 [7.42095757e-01 2.57904243e-01]
 [4.78294834e-03 9.95217052e-01]
 [9.99063869e-01 9.36130513e-04]
 [1.14088889e-01 8.85911111e-01]
 [9.64096408e-01 3.59035917e-02]
 [2.02749432e-01 7.97250568e-01]
 [1.86747246e-01 8.13252754e-01]
 [2.13815975e-01 7.86184025e-01]
 [2.90404175e-01 7.09595825e-01]
 [6.54029856e-01 3.45970144e-01]
 [7.96728512e-01 2.03271488e-01]
```

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